

**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY COMPUTER
SCIENCE & IT DEPARTMENT**

**Numerical Analysis (MS-203T)
Assignment 3
Sections: A B C D & E**

Semester: VI

Max Marks:04

Instructions:

- ☐ Attempt all questions

Question 1: Write a python code of Bisection method and display its output

```
def f(x):  
    return x**3 - x - 2  
  
def bisection(a, b, tol):  
    if f(a) * f(b) >= 0:  
        print("Invalid initial interval.")  
        return None  
  
    c = a  
    while (b - a) >= tol:  
        c = (a + b) / 2  
        if f(c) == 0.0:  
            break  
        elif f(c) * f(a) < 0:  
            b = c  
        else:  
            a = c  
        print(f"Midpoint: {c:.6f}, f(c): {f(c):.6f}")  
    return c  
  
root = bisection(1, 2, 0.0001)  
print(f"\nRoot using Bisection Method: {root:.6f}")
```

```
Midpoint: 1.500000, f(c): -0.125000  
Midpoint: 1.750000, f(c): 1.609375  
Midpoint: 1.625000, f(c): 0.666016  
Midpoint: 1.562500, f(c): 0.252197  
Midpoint: 1.531250, f(c): 0.059113  
Midpoint: 1.515625, f(c): -0.034054  
Midpoint: 1.523438, f(c): 0.012250  
Midpoint: 1.519531, f(c): -0.010971  
Midpoint: 1.521484, f(c): 0.000622  
Midpoint: 1.520508, f(c): -0.005179  
Midpoint: 1.520996, f(c): -0.002279  
Midpoint: 1.521240, f(c): -0.000829  
Midpoint: 1.521362, f(c): -0.000103  
Midpoint: 1.521423, f(c): 0.000259
```

Root using Bisection Method: 1.521423

Question 2: Write a python code of Newton Raphson method and display its output

```
def f(x):  
    return x**3 - x - 2  
  
def df(x):  
    return 3*x**2 - 1  
  
def newton_raphson(x0, tol, max_iter):  
    for i in range(max_iter):  
        x1 = x0 - f(x0)/df(x0)  
        print(f"Iteration {i+1}: x = {x1:.6f}")  
        if abs(x1 - x0) < tol:  
            return x1  
        x0 = x1  
    return x0  
  
root = newton_raphson(1.5, 0.0001, 10)  
print(f"\nRoot using Newton-Raphson Method: {root:.6f}")  
  
Iteration 1: x = 1.521739  
Iteration 2: x = 1.521380  
Iteration 3: x = 1.521380
```

Root using Newton-Raphson Method: 1.521380

Question 3: Write a python code of Iteration method and display its output

```
def g(x):  
    return (x + 2 / x**2) / 2 # From  $x^3 = x + 2$   
  
def fixed_point(x0, tol, max_iter):  
    for i in range(max_iter):  
        x1 = g(x0)  
        print(f"Iteration {i+1}: x = {x1:.6f}")  
        if abs(x1 - x0) < tol:  
            return x1  
        x0 = x1  
    return x0  
  
root = fixed_point(1.5, 0.0001, 20)  
print(f"\nRoot using Iteration Method: {root:.6f}")  
  
Iteration 1: x = 1.194444  
Iteration 2: x = 1.298142  
Iteration 3: x = 1.242482  
Iteration 4: x = 1.269009  
Iteration 5: x = 1.255474  
Iteration 6: x = 1.262168  
Iteration 7: x = 1.258804  
Iteration 8: x = 1.260481  
Iteration 9: x = 1.259641  
Iteration 10: x = 1.260061  
Iteration 11: x = 1.259851  
Iteration 12: x = 1.259956  
Iteration 13: x = 1.259904
```

Root using Iteration Method: 1.259904